Topology-Aware Retinal Vessel Segmentation Under Dataset Shift: A Controlled Study of TG-LiteNet

Abhiraj Agarwal Roll No. 2510110002
Shashwat Rajan Roll No. 2510111019
Tanish Shitanshu Roll No. 2510110008

CSD361 – Introduction to Machine Learning, Final Project Report
Department of Computer Science and Engineering, Shiv Nadar Institution of Eminence, Delhi NCR, India

***Abstract*—Photographs of the back of the eye can show early signs of diseases such as diabetes. A computer program that outlines the blood vessels in these photographs should keep thin vessels *connected*, work on cameras it was never trained on, and run on modest hardware. We study these three goals together with *TG-LiteNet*, a small neural network trained with a connectivity (topology) loss, robustness augmentation and an optional uncertainty estimate. Every model is trained on one dataset (DRIVE) and tested on three others (STARE, CHASE_DB1, HRF). We compare five build-up stages (5 random seeds each), all eight combinations of the three ingredients, and several simple control experiments. TG-LiteNet has 0.21 M parameters (U-Net: 31.04 M) and reaches a Dice score of 0.756 on the unseen datasets (U-Net: 0.762). The topology loss more than halves the number of broken vessel pieces (75.0 to 31.7) and adds 0.007 clDice, at a cost of 0.009 Dice. However, simply deleting tiny fragments gives every model about 5 broken pieces, and when the decision threshold is tuned on validation images the topology-trained models have the best Dice (0.782 against 0.757 for the U-Net). The small network saves memory but not time: a plain U-Net with base width 16 is as accurate, as fast and better connected. The uncertainty estimate finds wrong pixels moderately well but does not reliably flag bad images. No test image is used for training, tuning or model selection; we did look at the test results while designing the analyses, so the test sets are held-out rather than untouched.**

***Keywords*—retinal vessel segmentation, topology, domain generalization, lightweight networks, uncertainty, DRIVE**

I. Introduction

Doctors look at the blood vessels in the retina (the back of the eye) to screen for diabetic eye disease, glaucoma and high blood pressure [1]. Measuring vessel width and branching requires an accurate *vessel map* of the photograph. Since U-Net [2], such maps are produced by neural networks trained on a few small public datasets: DRIVE [3], STARE [4], CHASE_DB1 [5] and HRF [6]. The usual score, Dice, measures how much the predicted vessels overlap the true ones. It hides a different kind of error: a vessel that is cut in two barely changes Dice, yet it breaks the connectivity that doctors rely on [7]–[9].

Motivation. A model for a rural screening camp has to satisfy three needs at once: (1) keep vessels *connected* (topology); (2) keep working when the camera or patient group changes (*dataset shift*), ideally without labelled images from the new camera [12], [13]; and (3) be small and fast enough for cheap hardware [14]. We call the drop in Dice between the training dataset and a new dataset the *gap*.

Objectives. (O1) Build a reproducible pipeline that trains on one dataset only. (O2) Measure what each ingredient (topology loss, robustness augmentation, small backbone) contributes to accuracy, connectivity, the gap and cost. (O3) Test whether an uncertainty estimate can flag unreliable predictions.

Contributions. (1) A staged ablation and a full $2\times2\times2$ factorial study (backbone $\times$ topology loss $\times$ robustness augmentation) with repeated random seeds. (2) One evaluation that reports overlap, connectivity, the gap, model size, speed and uncertainty quality together. (3) Control experiments that check whether the gains survive simple alternatives, such as deleting tiny fragments, tuning the threshold, or using a plain small U-Net. (4) Open code that regenerates every number. We do *not* claim a new loss, a new architecture family or state-of-the-art accuracy.

II. Related Work

clDice (our base paper). Shit et al. [7] thin both the prediction and the ground truth to one-pixel *centrelines* (skeletons). They then check how much of the predicted centreline lies inside the true vessels and how much of the true centreline lies inside the prediction; the combined score is clDice. They give a differentiable version, *soft-clDice*, that can be used as a training loss. cbDice [8] adds vessel radius information, and Skeleton Recall [9] is a cheaper alternative. None of these was tested for training on one dataset and testing on others.

Dataset shift. Several methods generate artificial vessel trees or styles [15], or use frequency, edge or graph information [16]–[18]. A benchmark [13] found that, given enough data, even a plain U-Net generalises well.

Small networks. Galdran et al. [12] showed that networks with only 34 K to 68 K parameters can be competitive, and later work reports similar results [14], [19]. These papers report Dice on other datasets but not connectivity or uncertainty.

Topology and uncertainty. Recent methods adapt topology-aware models to new datasets [11], [20]–[22], but they use large models, labelled target data or a single dataset. Topology-aware uncertainty [23] scores whole structures.

TABLE I

WHAT THE CLOSEST WORKS REPORT (✓: YES, ∼: PARTLY, –: NO, N/V: NOT VERIFIABLE). PROTOCOL: SO = TRAINED ON ONE DATASET ONLY, MS = SEVERAL TRAINING DATASETS, SD = ONE DATASET IN TOTAL, TTA = ADAPTED AT TEST TIME, FT = FINE-TUNED ON LABELLED TARGET DATA.

Work	Protocol	Topology loss	Topology metrics	Cost	Uncertainty
clDice, Skeleton Recall [7], [9]	–	✓	✓	∼	–
DGSSA, WaveRNet [15], [16]	MS	–	–	–	–
Small networks [12], [14], [19]	SO	∼	–	✓	–
TFFM [20]	SD	✓	✓	–	–
TopoTTA [21]	TTA	∼	✓	n/v	–
TopoLoRA-SAM, TAPDecoderFT [11], [22]	FT	✓	✓	∼	–
Topology-aware uncertainty [23]	–	–	∼	n/v	✓
TG-LiteNet (this work)	SO	✓	✓	✓	✓

Monte-Carlo (MC) dropout [24] is a cheap alternative whose quality is known to degrade under dataset shift [25].

Research gap. Table I summarises the closest works. Among the papers we reviewed, none evaluates a small model trained on one dataset for connectivity, total cost and uncertainty together, with repeated runs. We fill this gap with a *study*, not a new method.

III. PROBLEM FORMULATION AND RESEARCH QUESTIONS

Given a colour photograph of the retina, the task is to label every pixel inside the circular field of view (FOV) as vessel or background. Models are trained on DRIVE only. STARE, CHASE_DB1 and HRF are never used for training, for choosing settings or for picking the best checkpoint (we did look at their results while designing the analyses, see Sec. VII-A). We ask four questions:

- **RQ1 (topology).** Does the topology loss improve connectivity without losing too much overlap?
- **RQ2 (generalisation).** Does robustness augmentation shrink the gap between DRIVE and the unseen datasets?
- **RQ3 (cost).** What do we lose in accuracy and connectivity by using a network with under 2 million parameters, and what do we save?
- **RQ4 (uncertainty).** Does MC-dropout uncertainty point to wrong pixels and images, especially on unseen datasets?

IV. PROPOSED METHODOLOGY

A. Overview

Fig. 1 shows the pipeline. A photograph is resized so that its field of view has a fixed width, converted to an enhanced green-channel image, and passed through an encoder–decoder network. The network has a mask output and, during training only, a second output that predicts vessel centrelines. At test time the model can be run $T{=}20$ times with dropout switched on; the average is the prediction and the disagreement between runs is the uncertainty map.

B. Preprocessing

The cameras differ a lot in resolution (HRF images are about 6× wider than DRIVE images), so the same vessel is a different number of pixels wide in each dataset. Convolutional networks are very sensitive to this. We therefore resize every image so that its FOV is 527 pixels wide (the median DRIVE width); this uses no labels and no statistics of the target datasets. We then keep the green channel, which shows vessels best, apply CLAHE [26] (a local contrast enhancement), and standardise the brightness inside the FOV. The same steps are used for training and testing.

C. Networks

U-Net baseline [2]: four down-sampling steps, double $3{\times}3$ convolution blocks and a base width of 64 channels (≈31.04 M parameters).

TG-LiteNet backbone. Each $3{\times}3$ convolution is split into a depthwise $3{\times}3$ convolution followed by a pointwise $1{\times}1$ convolution, as in MobileNets [27], which needs far fewer parameters. The widths are $(w, 2w, 4w, 8w, 8w)$ with $w{=}16$; the decoder uses bilinear up-sampling and skip connections. A $1{\times}1$ layer outputs the vessel mask. A second $1{\times}1$ layer (the *skeleton head*, used only in training) predicts the vessel centrelines. Dropout ($p{=}0.1$) is used only in stage E, for the uncertainty estimate.

D. Training objective

The total loss is

$$\mathcal{L} = \mathcal{L}_{\mathrm{Dice}} + \lambda_1 \mathcal{L}_{\mathrm{BCE}} + \lambda_2 \mathcal{L}_{\mathrm{topo}} + \lambda_3 \mathcal{L}_{\mathrm{cons}}, \tag{1}$$

where $\mathcal{L}_{\mathrm{Dice}}$ and $\mathcal{L}_{\mathrm{BCE}}$ are the standard overlap and pixel-wise losses, and all terms are computed inside the FOV.

Topology term. Let $S(\cdot)$ be the (soft) skeleton of a mask, P the predicted vessel probabilities and G the ground truth. The share of the predicted centreline that lies on true vessels is $T_{\mathrm{prec}} = \sum S(P) \odot G / \sum S(P)$, and the share of the true centreline that is found is $T_{\mathrm{sens}} = \sum S(G) \odot P / \sum S(G)$. The loss

$$\mathcal{L}_{\mathrm{clDice}} = 1 - \frac{2\, T_{\mathrm{prec}} T_{\mathrm{sens}}}{T_{\mathrm{prec}} + T_{\mathrm{sens}}} \tag{2}$$

is small only when both are high. The skeleton head adds a loss on the predicted centreline, so $\mathcal{L}_{\mathrm{topo}} = \mathcal{L}_{\mathrm{clDice}} + \gamma \mathcal{L}_{\mathrm{skel}}$. We also tried two other topology losses, Skeleton Recall [9] and cbDice [8], as variants (Sec. VII); our versions are re-implementations and may differ numerically from the authors' code.

Consistency term. Each training patch is shown to the network twice, once clean and once with random changes (below). The consistency loss asks the two predictions to agree.

E. Robustness (DG) augmentation

All stages use the same basic augmentation: random crops, flips, rotations and rescaling. From stage C on, we also apply, during training only, **style changes** (uneven illumination, blur, noise and JPEG compression) and **shape changes** (a smooth elastic bending and small patches of missing intensity). We call this *DG augmentation* (domain generalisation). It is deliberately much simpler than methods that synthesise whole vessel trees [15]. Because CLAHE and standardisation come afterwards, simple brightness and contrast changes are largely cancelled out by design.

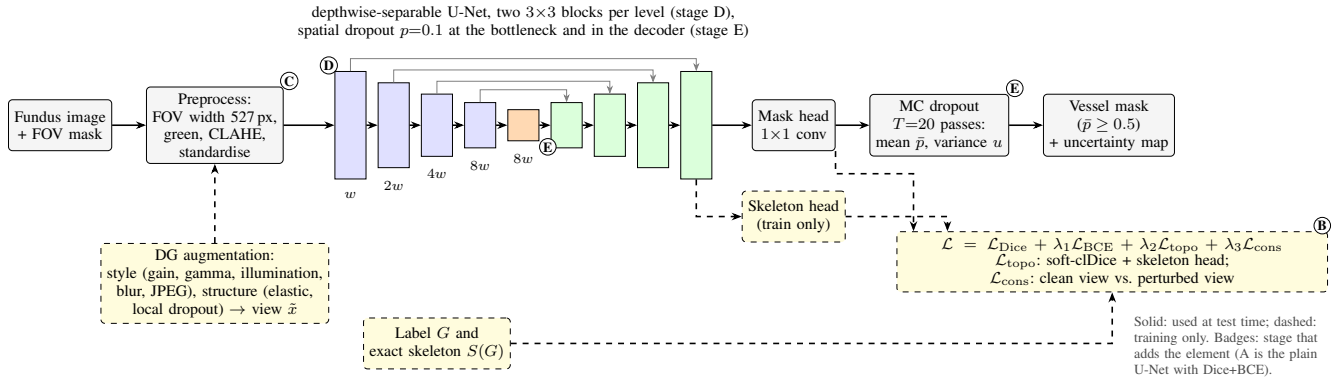


Fig. 1. TG-LiteNet. A photograph is resized to a fixed field-of-view width and passed through a small U-Net built from depthwise-separable convolutions. Repeated runs with dropout give the segmentation and an uncertainty map. The centreline output, the robustness augmentation and the loss (dashed) are used during training only.

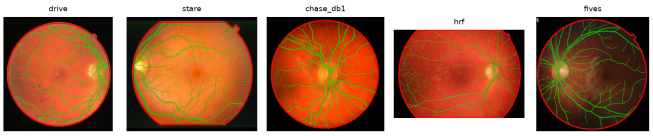


Fig. 2. The four datasets: vessel labels (green) and field-of-view outlines (red). Cameras, colours, fields of view and resolutions differ clearly; this is the dataset shift we study.

F. Uncertainty

With dropout switched on, we run the network $T=20$ times. The mean gives the final probability map and the variance of the 20 outputs gives a per-pixel uncertainty [24]. An image is scored by its average uncertainty. To reject unreliable cases, the pixels or images with the highest uncertainty are removed first.

V. Experimental Setup

A. Datasets

Table II lists the datasets. DRIVE has 20 training and 20 test images. We split the 20 training images once into 16 for training and 4 for validation; the validation images are the only ones used to pick checkpoints and settings. The 20 DRIVE test images are the in-domain test set. STARE (20 images), CHASE_DB1 (28) and HRF (45) are used for testing only, and we call them the *unseen* datasets. We use the first human annotator as ground truth. For HRF, and for DRIVE where available, the field-of-view masks are provided; for the other datasets we estimate them by thresholding and checked them visually (Fig. 2). About 9–13% of pixels are vessels in every dataset, so the classes are imbalanced (about 1:8).

The table shows how large the shift is: the average green-channel brightness ranges from about 60 (CHASE_DB1, HRF) to about 110 (STARE). After resizing, vessels are about 3.5 pixels wide in DRIVE, STARE and CHASE_DB1, but only about 1.4 pixels in HRF, because the HRF annotations are thin compared with its large images. HRF therefore also measures a difference in annotation style, not only a difference in camera.

TABLE II
DATASETS (ALL STATISTICS MEASURED ON THE FILES USED IN OUR EXPERIMENTS). FOV = FIELD-OF-VIEW WIDTH, USED TO NORMALISE THE SCALE OF EVERY IMAGE; CALIBRE = MEAN VESSEL WIDTH (2× DISTANCE TRANSFORM ON THE LABEL SKELETON) AT NATIVE RESOLUTION AND AFTER NORMALISATION TO A 527 PX FOV; G = GREEN-CHANNEL MEAN±STD INSIDE THE FOV (0–255). SOURCE = TRAIN (16) / VALIDATION (4) / TEST (20 OFFICIAL TEST IMAGES); TARGETS ARE TEST-ONLY.

Dataset	#	Size (px)	Vessel %	FOV (px)	Calibre (px)	G	Role
DRIVE [3]	40	565×584	12.9	527	3.5→3.5	97±17	source
STARE [4]	20	700×605	10.5	638	4.3→3.5	109±24	target
CHASE_DB1 [5]	28	999×960	10.3	899	6.1→3.6	62±26	target
HRF [6]	45	3504×2336	9.1	3262	8.8→1.4	60±17	target

B. Training procedure

The code uses Python and PyTorch [28] (2.11.0+cu128), with OpenCV and scikit-image [29] for image processing. Training ran on Kaggle (Tesla T4; 37 runs) and Colab (NVIDIA L4; 33 runs); the code versions differ only in analysis scripts and added experiments, so results are statistically, not bit-for-bit, reproducible. All stages share one optimiser and schedule (AdamW, learning rate 10^{-3}, cosine decay, 5000 iterations of 8 patches of 256×256 pixels; `configs/base.yaml`) and differ by one switch each:

- **A** plain U-Net (Dice + BCE loss);
- **B** A + topology loss;
- **C** B + DG augmentation and consistency loss;
- **D** C with the small depthwise backbone;
- **E** D + dropout, with MC averaging (this is *TG-LiteNet*).

Stages A–E use 5 random seeds each; every other configuration uses 3. We report the mean±standard deviation over seeds. When we compare two models we average the difference over the seeds they share and quote a 95% range "±"; a difference smaller than its ± is not clear evidence. These ranges are descriptive: we ran many comparisons and did not correct for that.

C. Metrics

All metrics are computed inside the FOV after thresholding the probability map at a fixed 0.5 (chosen before seeing any unseen data), against the ground truth at its *original* resolution

(the model's output is resized back, because shrinking thin labels would break them).

- **Dice**: overlap between predicted and true vessels (higher is better).
- **clDice**: the centreline score of Sec. IV (higher is better); it measures connectivity.
- **Betti-0 error**: the difference between the number of separate vessel pieces in the prediction and in the ground truth (lower is better). Stray fragments increase it.
- **Gap**: Dice on DRIVE minus Dice on unseen data. "Unseen" always means the average of the three unseen datasets.
- **Cost**: number of parameters, floating-point operations and time for one 512×512 image.
- **Uncertainty**: the rank correlation ρ between uncertainty and error, and risk–coverage curves (how the error falls when the most uncertain pixels or images are removed).

VI. Experimental Results and Discussion

A. Main results

Table III and Fig. 3 give the staged ablation.

Topology loss (A→B, RQ1). The topology loss cuts the unseen Betti-0 error from 75.0 to 31.7 (-43.3 ± 16.0, on every unseen dataset) and raises clDice by 0.007 (±0.004). It costs 0.009 unseen Dice (±0.004) and 0.003 on DRIVE (±0.002): Dice is lower for 92% of the 93 unseen images and clDice is higher for 96%. It does not reduce loop errors (Betti-1 changes only from 22.5 to 21.0). So RQ1 holds for the centreline score and the piece count, not for overlap or loops. The checks below show that the piece count is easy to fix by cleanup and that much of the Dice cost is a threshold effect.

Robustness augmentation (B→C, RQ2). DG augmentation raises unseen Dice by 0.008 (±0.005; higher for 81% of images) and shrinks the gap from 0.074 to 0.065 (-0.009 ± 0.004). On CHASE_DB1 the fraction of vessels found (sensitivity) rises from 0.879 to 0.927. Connectivity changes little (clDice $+0.004\pm0.004$). RQ2 holds for overlap and only weakly for connectivity.

Small backbone (C→D→E, RQ3). Going from 31 M to 0.21 M parameters costs 0.004 Dice on DRIVE (±0.002), 0.005 unseen Dice (±0.003) and clearly more connectivity: unseen clDice drops by 0.014 (±0.004) and the Betti-0 error rises by 16.8 (±10.3), mostly on HRF, whose thin annotations reward a bigger network (sensitivity 0.891 to 0.862). The gap stays at 0.065. Adding dropout (E) changes Dice by at most 0.002 and does not change clDice or Betti-0, so dropout gives only the uncertainty map. Overall (A→E), unseen Dice is 0.007 lower (±0.007) and clDice 0.003 lower (±0.003).

What kind of mistakes? On unseen data all stages find most vessels (recall 0.90–0.92) but also mark many pixels that are not vessels (precision 0.66–0.67, against 0.81 on DRIVE). False positives (4.5–4.8% of the FOV) are about five times as common as false negatives (0.8–1.0%). The typical error is a spurious thin vessel or a stray fragment, not a missed vessel.

B. Three checks that change the story

These checks need no retraining. Fig. 4 shows the first two (stages A–E: 5 seeds; DG-only U-Net: 3 seeds).

1. Deleting tiny fragments. The ground truth has 3–12 vessel pieces per image, the predictions 25–187. Removing pieces smaller than a size chosen on the DRIVE validation images (100–200 pixels at our scale) brings the unseen Betti-0 error of *every* model to 4.6–5.0, including the plain U-Net (75.0 to 4.7), while Dice rises by at most 0.002 and clDice falls by at most 0.004. So the piece-count advantage of the topology loss is not specific to it; a one-line cleanup gives the same result. After cleanup the clDice gain is $+0.006$ (±0.008, not clear) for topology alone and $+0.012$ (±0.005) for topology plus DG augmentation.

2. Choosing the threshold. The topology-trained models over-segment at the default threshold of 0.5 (recall 0.90, precision 0.66). If the threshold is instead chosen on the four DRIVE validation images, it becomes 0.50–0.65 for these models (0.45–0.50 for the plain U-Net), and their unseen Dice rises by 0.015–0.020 (B 0.771, C 0.782, D 0.772, E 0.771) while the plain U-Net does not improve (-0.006 ± 0.007). At tuned thresholds the topology loss therefore *gains* Dice instead of losing it (A→B $+0.014\pm0.008$, A→C $+0.025\pm0.006$): the Dice cost at 0.5 is mostly a calibration effect. A U-Net with DG augmentation but no topology loss reaches 0.777, and stage C is $+0.002\pm0.002$ above it (p=0.05, 3 seeds) with 0.007 more clDice. Because the thresholds come from only four images and were examined after the fact, all headline results keep 0.5.

3. Image scale. If images are *not* resized to a fixed FOV width (seeds 3–4), the plain U-Net collapses (unseen Dice 0.768 to 0.637, HRF 0.50), while stages D and E hold up (0.757 to 0.750 and 0.756 to 0.751). Only A, D and E were run this way, so we cannot say why the small networks are more robust.

C. Efficiency and the trade-off (RQ3)

TG-LiteNet has 147× fewer parameters and 70× fewer operations than the U-Net, but it is only 6.8–8.8× faster on a CPU, depending on the machine (laptop: 209 against 1834 ms; Kaggle: 429 against 2915 ms; T4 GPU: 12.2 against 91.1 ms). Depthwise convolutions do little work per byte of memory read, so operation counts overstate the speed-up. Against *plain* small U-Nets the picture reverses (Fig. 5, Table V): a base-8 U-Net with the same number of operations runs in 67 ms, 3.1× faster than TG-LiteNet, with the same Dice; a base-16 U-Net (1.94 M parameters, still under 2 M) is as fast (187 ms) and has better clDice and Betti-0. The depthwise backbone saves parameters, not time, so for CPU-only hardware a base-16 U-Net is the better choice on this evidence. Compared with the U-Net, TG-LiteNet gives up 0.008 DRIVE Dice, 0.006 unseen Dice and 0.003 unseen clDice; the loss is no larger on unseen data (gap 0.065 for both C and D). MC dropout multiplies the cost by 20.

TABLE III

MAIN RESULTS (MEAN±STD OVER RANDOM SEEDS). DICE AND CLDICE: HIGHER IS BETTER; BETTI-0 ERROR AND GAP: LOWER IS BETTER. UNSEEN = AVERAGE OF STARE, CHASE_DB1 AND HRF; GAP = DICE ON DRIVE − DICE ON UNSEEN DATA. CPU TIME IS FOR ONE 512×512 IMAGE ON FOUR LAPTOP CPU THREADS. THE BEST VALUE IN EACH COLUMN IS IN BOLD.

	DRIVE	Unseen datasets					
Model	Dice	Dice	clDice	Betti-0 err.	Gap	Params (M)	CPU (ms)
A U-Net	**0.830**±0.001	**0.762**±0.006	0.835±0.002	75.0±14.4	0.068±0.006	31.04	1834
B + topology loss	0.827±0.001	0.753±0.004	0.843±0.003	31.7±2.3	0.074±0.004	31.04	1834
C + DG augmentation	0.826±0.001	0.761±0.001	**0.847**±0.001	**27.9**±2.1	**0.065**±0.001	31.04	1834
D + compact backbone	0.822±0.001	0.757±0.003	0.832±0.002	44.7±6.5	0.065±0.003	0.21	209
E TG-LiteNet	0.822±0.001	0.755±0.005	0.833±0.002	41.9±2.8	0.066±0.005	0.21	209
E with MC dropout (20 passes)	0.822±0.001	0.756±0.004	0.833±0.002	42.7±3.0	0.066±0.004	0.21	209 × 20

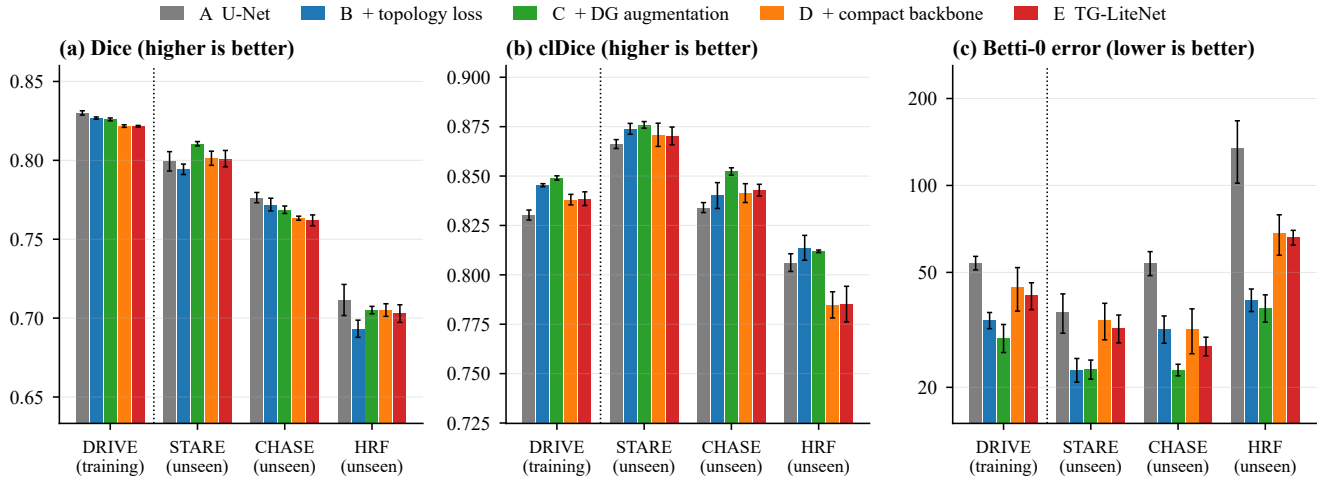


Fig. 3. Results of stages A–E on each dataset (mean over seeds; whiskers: standard deviation over seeds). Betti-0 error is on a log scale.

D. Uncertainty (RQ4)

At pixel level the MC-dropout uncertainty is moderately informative: its rank correlation with the errors is 0.31 on DRIVE and slightly higher on the unseen datasets (0.32 STARE, 0.34 CHASE_DB1, 0.34 HRF), so the expected increase under shift is present but small. Removing the 10% most uncertain pixels (Fig. 6a) lowers the pixel error from 4.7% to 1.6% on DRIVE and from 6.5% to 3.5% on HRF. At image level (Fig. 6b) the evidence is weak. Removing the most uncertain images is no better than removing random images on DRIVE and STARE, 4% better on CHASE_DB1 (correlation 0.48) and *worse* than random on HRF (correlation −0.17). With only 20–45 images per dataset these numbers are imprecise, but they do not support using this uncertainty to refer whole images [25]. Together with the 20× cost, they do not justify MC dropout here; deep ensembles or temperature scaling (not run) would be the next options. We also did not measure calibration.

E. Qualitative results

Fig. 7 shows one image per dataset (seed 0). The windows were chosen where stages A and E differ most, so differences look larger than they are on average. The errors of all three models are mostly thin red (false positive) or blue (missed) fragments along the finest vessels; the large vessels are almost error-free, and some spurious red segments, for example on CHASE_DB1, appear in all three models. On HRF the small models miss more fine vessels than the U-Net (average HRF sensitivity 0.864 for E and 0.889 for A), which matches their lower clDice there. The uncertainty maps of E mark vessel borders and thin segments in general rather than the specific errors, in line with the weak image-level result.

TABLE IV

ALL EIGHT COMBINATIONS OF BACKBONE (U-NET OR COMPACT), TOPOLOGY LOSS AND DG AUGMENTATION. UNSEEN = AVERAGE OF THE THREE UNSEEN DATASETS; GAP = DICE ON DRIVE − DICE ON UNSEEN DATA. ROWS 1, 4, 3 AND 8 ARE STAGES A, C, B AND D (5 SEEDS); THE OTHER ROWS HAVE 3 SEEDS.

Backbone	Topology	DG	DRIVE Dice	Unseen Dice	Unseen clDice	Gap
U-Net	–	–	0.830±0.001	0.762±0.006	0.835±0.002	0.068±0.006
U-Net	–	✓	0.831±0.000	0.771±0.001	0.837±0.000	0.060±0.001
U-Net	✓	–	0.827±0.001	0.753±0.004	0.843±0.003	0.074±0.004
U-Net	✓	✓	0.826±0.001	0.761±0.001	0.847±0.001	0.065±0.001
Lite	–	–	0.823±0.001	0.745±0.011	0.824±0.001	0.078±0.011
Lite	–	✓	0.824±0.000	0.767±0.002	0.827±0.001	0.057±0.002
Lite	✓	–	0.821±0.002	0.747±0.007	0.829±0.004	0.074±0.009
Lite	✓	✓	0.822±0.001	0.757±0.003	0.832±0.002	0.065±0.003

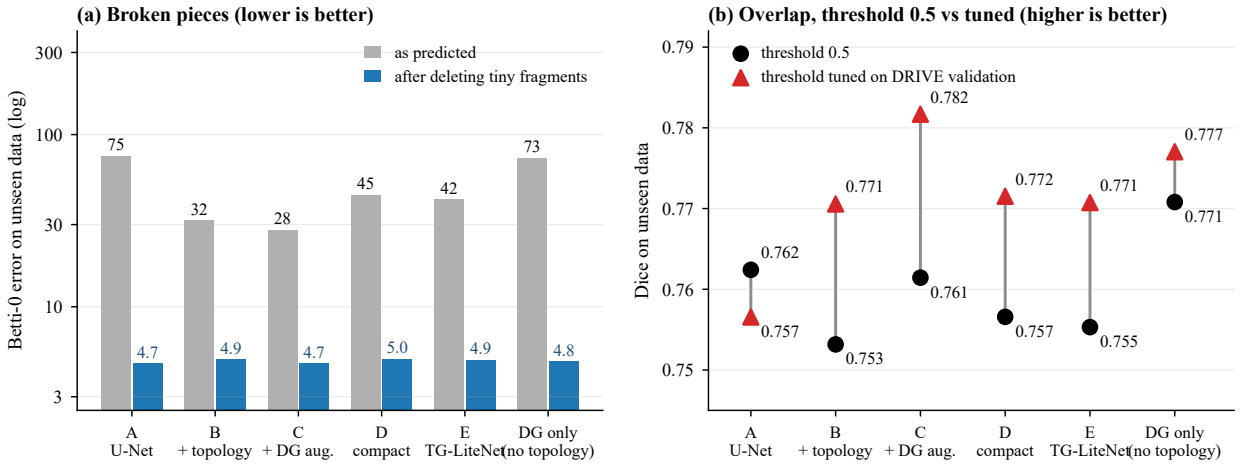


Fig. 4. Two checks on the unseen datasets. (a) Betti-0 error as predicted (grey) and after deleting fragments smaller than a size chosen on DRIVE validation (blue); the number above each bar is its value. (b) Dice at the standard threshold 0.5 (black) and at the threshold tuned on DRIVE validation (red).

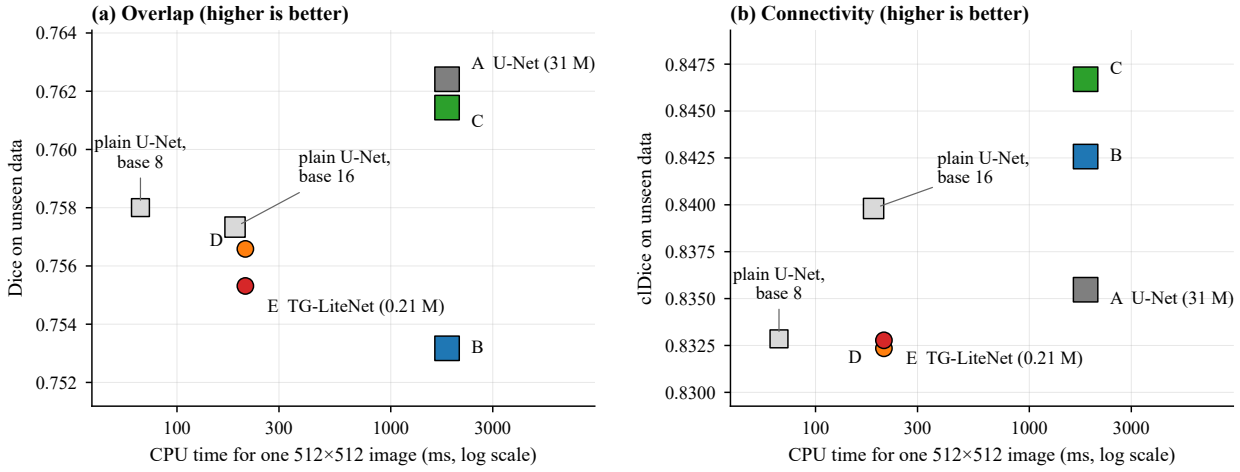


Fig. 5. Accuracy against CPU time (laptop benchmark) for the main models. Bigger markers mean more parameters; squares are U-Nets and circles are TG-LiteNet variants. Better models are higher and further left.

VII. Ablation Analysis

Table IV tries all eight combinations of backbone, topology loss and DG augmentation. The *topology loss* lowers the raw Betti-0 error in all four pairs (U-Net 75→32 and 73→28, small network 86→46 and 88→45), raises clDice by 0.005–0.009 and changes unseen Dice by $+0.002$ to -0.011; so it still helps a network under 2 M parameters, with the cleanup caveat above. *DG augmentation* adds 0.008 to 0.022 unseen Dice, narrows the gap (0.068→0.060 for the plain U-Net) and leaves Betti-0 unchanged. The *small backbone* loses 0.004–0.013 Dice and 0.011–0.014 clDice and adds 15–20 Betti-0 error in every combination. At threshold 0.5, the best unseen Dice comes from DG augmentation without topology loss (U-Net 0.771; small network 0.767, $+0.009 \pm 0.006$ over the baseline but with worse clDice, -0.011 ± 0.006). The full TG-LiteNet recipe is thus an operating point that favours connectivity.

Table V (top) shows that the default soft-clDice has the best Dice: cbDice and clDice+Skeleton Recall lower the Betti-0

TABLE V

Variants of stage D on unseen data: other topology losses (top); removed ingredients and plain compact U-Nets trained the same way (middle); network width (bottom). Stage D has 5 seeds, other rows 3. CPU time: 4 laptop threads, one 512×512 image.

	Dice	clDice	Betti-0 err.	Params (M)	CPU (ms)
soft-clDice + head (D)	0.757±0.003	0.832±0.002	44.7	0.21	209
cbDice	0.745±0.002	0.834±0.002	39.1	0.21	209
Skeleton Recall	0.743±0.006	0.828±0.001	55.3	0.21	209
clDice + Skel. Recall	0.734±0.001	0.832±0.003	36.5	0.21	209
D without head	0.761±0.006	0.831±0.001	53.8	0.21	209
D without consistency	0.759±0.005	0.833±0.002	52.2	0.21	209
plain U-Net, base 8	0.758±0.006	0.833±0.001	48.9	0.49	67
plain U-Net, base 16	0.757±0.005	0.840±0.001	33.7	1.94	187
width w=8	0.754±0.005	0.822±0.001	61.1	0.06	76
width w=16 (D)	0.757±0.003	0.832±0.002	44.7	0.21	209
width w=24	0.764±0.002	0.835±0.003	40.0	0.46	343
width w=32	0.759±0.005	0.838±0.003	34.3	0.81	487
width w=48	0.759±0.005	0.839±0.002	30.1	1.81	838

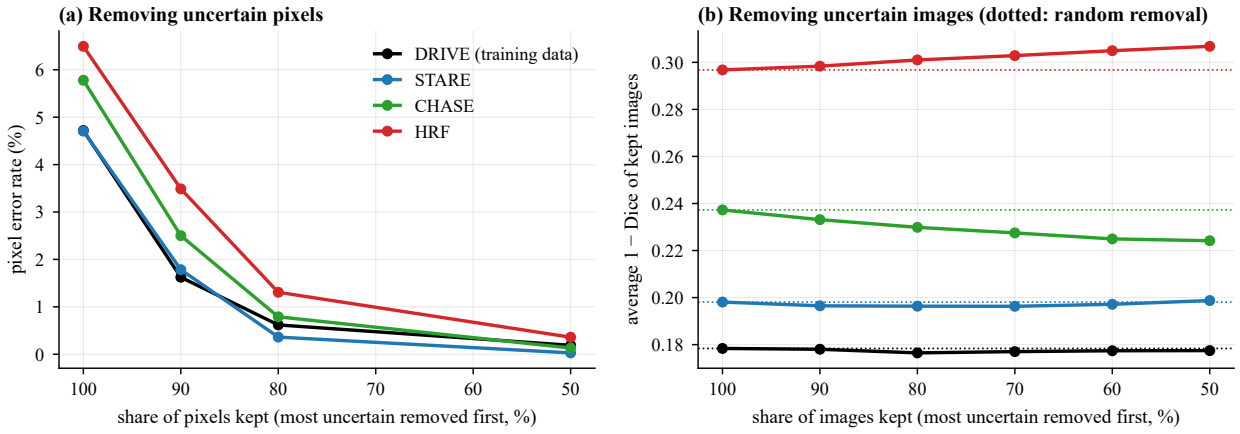


Fig. 6. Risk–coverage of the MC-dropout uncertainty of TG-LiteNet. (a) Pixel error after removing the most uncertain pixels. (b) Average 1−Dice of the images that are kept when the most uncertain images are removed; dotted lines show random removal. A useful uncertainty would make the solid lines fall.

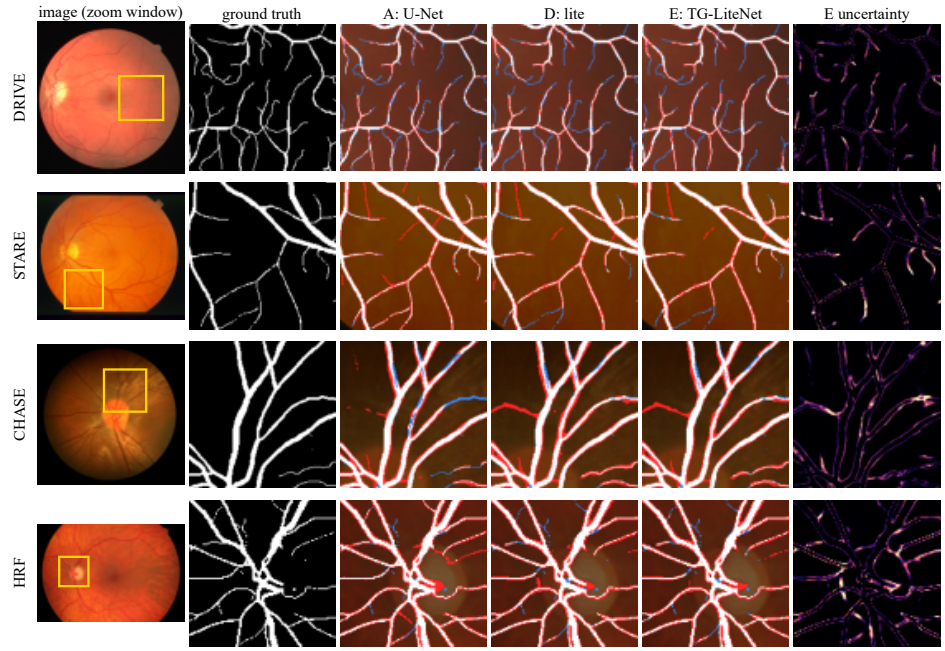


Fig. 7. Examples (seed 0, one image per dataset). The yellow window is enlarged to the right: ground truth, error maps of A, D and E (white: correct, red: false positive, blue: missed; threshold 0.5) and the MC-dropout uncertainty of E.

error a little (44.7 to 39.1 and 36.5, within the seed spread) at a cost of 0.012 ($\pm$0.013) and 0.023 ($\pm$0.011) Dice. Neither the skeleton head nor the consistency loss improves Dice or clDice (without the head: 0.761 and 0.831; without consistency: 0.759 and 0.833; stage D: 0.757 and 0.832); the only trace of a benefit is a lower Betti-0 error (10 $\pm$ 3 and 8 $\pm$ 9 lower). Plain small U-Nets trained the same way match the Dice of the small network (base 8, 0.49 M parameters: 0.758; base 16, 1.94 M: 0.757), and the larger one is better connected (clDice 0.840 against 0.832, difference $+0.006 \pm 0.003$; Betti-0 33.7 against 44.7). In the width sweep, Dice stops improving near w=24, while clDice (0.822 to 0.839) and Betti-0 (61 to 30) keep improving up to w=48.

A. *Limitations*

(i) There are only 16 training images, and seed-to-seed variation can match small gains (stage A's unseen Dice ranges from 0.755 to 0.769 over five seeds). Stages A–E use 5 seeds, other configurations 3 and the image-scale check 2. (ii) The unseen average weights three very different datasets equally, and raw Betti counts are dominated by HRF. (iii) The image-scale check covers only A, D and E. (iv) Labels come from one annotator, and four of five field-of-view sets are estimated. (v) Soft-clDice, Skeleton Recall and cbDice are our own re-implementations. (vi) Speed was measured with standard

PyTorch on a laptop, not on an edge device, and varied up to 2× on shared cloud machines. (vii) The unseen datasets were never used for training or tuning, but we saw their results while designing the experiments, so they are held-out rather than untouched; comparisons are uncorrected, and the fragment size and tuned thresholds (four validation images) were chosen after the fact. (viii) The DG stage bundles augmentation, a consistency loss and a second view; the controls remove only the consistency loss and the skeleton head. (ix) Uncertainty is ranked, not calibrated. (x) Training on one dataset only is harder than using several datasets or adapting at test time, so our numbers cannot be compared with such papers [15], [16], [21]. (xi) Checkpoints are not in the repository, so the committed per-image results can be reproduced by retraining but not by re-evaluating.

VIII. Conclusion and Future Work

We built and tested TG-LiteNet with models trained on DRIVE only and one fixed threshold of 0.5 unless stated. Our answers to the research questions are:

- **RQ1, topology.** The topology loss cuts the unseen Betti-0 error of the U-Net from 75 to 32 and raises clDice by 0.007, at 0.009 Dice. But deleting tiny fragments gives every model a Betti-0 error of about 5, so the lasting benefit is the small clDice gain. With a threshold tuned on DRIVE validation, the topology-trained U-Nets have the highest Dice (stage C 0.782, plain U-Net 0.757), so the Dice cost at 0.5 is mostly calibration.
- **RQ2, generalisation.** DG augmentation adds 0.008 Dice and narrows the gap from 0.074 to 0.065. At threshold 0.5 the best Dice comes from DG without topology loss (0.771), but at tuned thresholds stage C is as good or slightly better (0.782 against 0.777) and better connected.
- **RQ3, cost.** TG-LiteNet is 0.006 Dice below the U-Net with 147× fewer parameters. However, plain small U-Nets match its accuracy (base 8) or also its connectivity (base 16, clDice 0.840) at equal or lower CPU time, so a base-16 U-Net is the better deployment choice on this evidence. The skeleton head, the consistency loss and dropout add nothing measurable.
- **RQ4, uncertainty.** MC dropout ranks pixel errors moderately (correlation 0.31–0.34) but is within 4% of random at image level and worse on HRF, so at 20× the cost we would drop it.

The **limitations** are listed in Sec. VII-A. **Future work:** a gradual increase of the topology weight during training, deep ensembles or temperature scaling with calibration metrics, the official cbDice code, evaluation on the held-out FIVES dataset, and speed measurements on a real edge device.

References

[1] Z. Liu, M. S. Sunar, T. S. Tan, and W. H. Wan Hitam, "Deep learning for retinal vessel segmentation: a systematic review of techniques and applications," *Med. Biol. Eng. Comput.*, vol. 63, no. 8, pp. 2191–2208, 2025.

[2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. MICCAI*, 2015, pp. 234–241.

[3] J. Staal, M. D. Abràmoff, M. Niemeijer, M. A. Viergever, and B. van Ginneken, "Ridge-based vessel segmentation in color images of the retina," *IEEE Trans. Med. Imag.*, vol. 23, no. 4, pp. 501–509, 2004.

[4] A. Hoover, V. Kouznetsova, and M. Goldbaum, "Locating blood vessels in retinal images by piecewise threshold probing of a matched filter response," *IEEE Trans. Med. Imag.*, vol. 19, no. 3, pp. 203–210, 2000.

[5] M. M. Fraz *et al.*, "An ensemble classification-based approach applied to retinal blood vessel segmentation," *IEEE Trans. Biomed. Eng.*, vol. 59, no. 9, pp. 2538–2548, 2012.

[6] A. Budai, R. Bock, A. Maier, J. Hornegger, and G. Michelson, "Robust vessel segmentation in fundus images," *Int. J. Biomed. Imaging*, vol. 2013, 2013.

[7] S. Shit *et al.*, "clDice — a novel topology-preserving loss function for tubular structure segmentation," in *Proc. IEEE/CVF CVPR*, 2021, pp. 16 560–16 569.

[8] P. Shi, J. Hu, Y. Yang, Z. Gao, W. Liu, and T. Ma, "Centerline boundary Dice loss for vascular segmentation," in *Proc. MICCAI*, LNCS 15008, 2024, pp. 46–56.

[9] Y. Kirchhoff *et al.*, "Skeleton recall loss for connectivity conserving and resource efficient segmentation of thin tubular structures," in *Proc. ECCV*, LNCS 15135, 2024, pp. 218–234.

[10] X. Hu, F. Li, D. Samaras, and C. Chen, "Topology-preserving deep image segmentation," in *Proc. NeurIPS*, 2019.

[11] Y. Huang, T. Miyazaki, K. Xu, X. Liu, Y. Fan, and S. Omachi, "Topology-aware parameter-efficient adaptation for cross-dataset retinal vessel segmentation," arXiv:2609.26189, 2026.

[12] A. Galdran *et al.*, "State-of-the-art retinal vessel segmentation with minimalistic models," *Sci. Rep.*, vol. 12, art. 6174, 2022.

[13] J. Fadugba, P. Köhler, L. Koch, P. Manescu, and P. Berens, "Benchmarking retinal blood vessel segmentation models for cross-dataset and cross-disease generalization," arXiv:2406.14994, 2024.

[14] S. Sobhan and F. Jalil, "LightVesselNet: An ultra-lightweight sub-100K parameter network for retinal blood vessel segmentation," arXiv:2606.05354, 2026.

[15] B. Liu, Y. Zhang, S. Wang, S. Li, and J. Hong, "DGSSA: Domain generalization with structural and stylistic augmentation for retinal vessel segmentation," *Neural Netw.*, vol. 194, art. 108118, 2026.

[16] C. Wang, Y. Wang, Q. Xu, and G. Chen, "WaveRNet: Wavelet-guided frequency learning for multi-source domain-generalized retinal vessel segmentation," arXiv:2601.05942, 2026.

[17] D. Hu, H. Li, H. Liu, and I. Oguz, "Domain generalization for retinal vessel segmentation via Hessian-based vector field," *Med. Image Anal.*, vol. 95, art. 103164, 2024.

[18] K. Liu, S. Gao, Y. Fu, and S. Gao, "Towards generalizable retina vessel segmentation with deformable graph priors," in *Proc. NeurIPS*, 2025.

[19] Z. Zeng, J. Liu, X. Huang, K. Luo, X. Yuan, and Y. Zhu, "Efficient retinal vessel segmentation with 78K parameters," *J. Imaging*, vol. 11, no. 9, art. 306, 2025.

[20] I. Ahmed, S. Absar, A. A. Sami, S. Sakib, D. Biswas, and S. A. M. Mostafa, "TFFM: Topology-aware feature fusion module via latent graph reasoning for retinal vessel segmentation," arXiv:2601.19136, 2026.

[21] J. Zhou *et al.*, "TopoTTA: Topology-enhanced test-time adaptation for tubular structure segmentation," in *Proc. IEEE/CVF ICCV*, 2025, pp. 24 123–24 134.

[22] S. Khazem, "TopoLoRA-SAM: Topology-aware parameter-efficient adaptation of foundation segmenters for thin-structure and cross-domain binary semantic segmentation," arXiv:2601.02273, 2026.

[23] S. Gupta, Y. Zhang, X. Hu, P. Prasanna, and C. Chen, "Topology-aware uncertainty for image segmentation," in *Proc. NeurIPS*, 2023.

[24] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: Representing model uncertainty in deep learning," in *Proc. ICML*, 2016, pp. 1050–1059.

[25] Y. Ovadia *et al.*, "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Proc. NeurIPS*, 2019.

[26] K. Zuiderveld, "Contrast limited adaptive histogram equalization," in *Graphics Gems IV*. Academic Press, 1994, pp. 474–485.

[27] A. G. Howard *et al.*, "MobileNets: Efficient convolutional neural networks for mobile vision applications," arXiv:1704.04861, 2017.

[28] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Proc. NeurIPS*, 2019.

[29] S. van der Walt *et al.*, "scikit-image: image processing in Python," *PeerJ*, vol. 2, art. e453, 2014.

[30] T. Y. Zhang and C. Y. Suen, "A fast parallel algorithm for thinning digital patterns," *Commun. ACM*, vol. 27, no. 3, pp. 236–239, 1984.